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标题	Hydrogenation of Lactic Acid to 1,2 - Propanediol over Ru - Based Catalysts
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摘要（原文节选）

AbstractThe catalytic hydrogenation of lactic acid to 1,2 - propanediol with supported Ru catalysts in water was investigated. The influence of catalyst support (activated carbon, γ - Al2O3, SiO2, TiO2, and CeO2) and promoters (Pd, Au, Mo, Re, Sn) on the catalytic performance was evaluated. Catalytic tests revealed that TiO2 yields the best Ru catalysts. With a monometallic Ru/TiO2 catalyst, a 1,2 - propanediol yield of 70 % at 79 % lactic acid conversion was achieved at 130 ° C after 20 h reaction. Minor byproducts of th ⋯

数据表 · Table 1（7 行，含数值行 7 行）

Overview of LA hydrogenation over Ru (5 wt. %) based catalyst in aqueous solution. C 0(LA), initial LA concentration; X(LA), conversion in %; S(1,2 - PDO), selectivity to 1,2 - PDO in %.

Catalyst	C 0(LA)[m]	P H2 [bar]	T [° C]	t [h]	X(LA)[%]	S(1,2 - PDO)[%]	Ref.
Ru/SiO2	1	80	130	7	30	80	19
Ru/C	1	50	130	2	70	84	20
RuSn/C	1.66	60	190	4	>95	>95	21
Ru/C	0.55	35	120	2.5	19	>95	22
RuMoOx/C	0.55	80	120	18	>95	95	23
Ru/C	0.55	80	120	2	13	93	23
Ru/SiO2	0.55	80	120	2	4.7	92	23

正文对应句（逐字）：The activity of the supported Ru catalysts was evaluated in the batch hydrogenation of aqueous 0.1 m LA at 130 ° C. LA conversion and product yield obtained after 4 h are displayed in Figure 3. Ru/TiO2 gave the highest 1,2 - PDO yield (27 %) at a LA conversion of 39 %. The Ru/SiO2, Ru/C, and Ru/Al2O3 c

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